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Homology modeling in the twilight zone: Improved accuracy by sequence space analysis

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Abstract

The analysis of the relationship between sequence and structure similarities during the evolution of a protein family has revealed a limit of sequence divergence for which structural conservation can be confidently assumed and homology modeling is reliable. Below this limit, the twilight zone corresponds to sequence divergence for which homology modeling becomes increasingly difficult and requires specific methods. Either with conventional “threading” methods or with recent deep-learning methods, such as AlphaFold, the challenge relies on the identification of a template that shares not only a common ancestor (homology) but also a conserved structure with the query. As both homology and structural conservation are transitive properties, mining of sequence databases followed by multidimensional scaling (MDS) of the query sequence space can reveal intermediary sequences to infer homology and structural conservation between the query and the template. Here, as a case study, we studied the plethodontid receptivity factor isoform 1 (PRF1) from *Plethodon jordani*, a member of a pheromone protein family present only in lungless salamanders and weakly related to cytokines of the IL6 family. A variety of conventional threading methods led to the cytokine CNTF as a template. Sequence mining, followed by phylogenetic and MDS analysis, provides missing links between PRF1 and CNTF and allows reliable homology modeling. In addition, we compare automated models obtained from web servers to a customized model to show how modeling can be improved by expert information.

1 Introduction

Since the resolution of the myoglobin structure in 1958 [1], the number of protein structures deposited in the Protein Data Bank [2] has increased exponentially to reach more than 160 000 structures in 2020. These structures led to a better understanding of protein functions and mechanisms of action. They have paved the way to computational approaches for rational drug design, search of targetable allosteric sites, better understanding of structure-function relationships, and so on. However, in spite of the huge advances in the field, the sequence space increases much more than structural space and computational approaches towards many proteins still rely on molecular modeling.

Presently, based on available structural information and deposited structures, proteins (or protein regions) can be classified into four categories: (1) proteins with resolved structures, (2) proteins with

closely related structurally resolved homologs that can be straightforwardly modeled by homology, (3) proteins within the twilight zone, for which structural information can be reached in spite of the absence of close structurally resolved homologs, and (4) proteins within the dark zone that lack similarity to any known structure and are inaccessible to homology modeling [3]. A recent analysis of the human genome, carried out with the deep-learning based AlphaFold program, revealed that on a residue basis, 58% of total residues have a resolved structure or are modelled with high confidence, whereas about 20% are dark and the remaining ones are in the twilight zone [4].

The initial studies from the 1990s on the relationship between sequence and structural evolution remain valid today. In a hallmark paper, Sander and Schneider [5] determined a curve describing the limit of confidence between evolution of the sequence and conservation of the structure. This study was extended by Rost [6] who corroborated the main result: a length dependent cutoff line separates close homologs with structural conservation from remote homologs with unknown structural similarity. The length of the aligned sequence is a key factor for confidence level in structural conservation. A cutoff of about 20% for a sequence of 200 amino acids or longer separates the safe zone for homology modeling from the twilight zone (Fig. 1). This cutoff does not mean that homology modeling is not possible in the twilight zone, but it points to the additional difficulties that arise in the twilight zone.

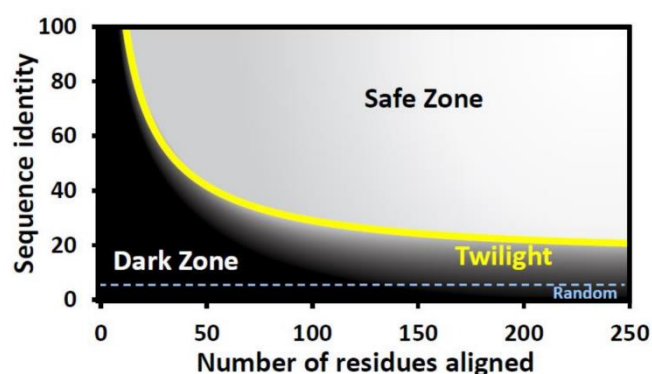


Fig. 1: Schematic representation of the dark, twilight and safe zones for molecular modeling of a protein as a function of the sequence identity and aligned length between the query and the template. Above the yellow line (drawn from [5]), the light grey zone indicates the safe zone of homology modeling. Below the yellow zone, the dark grey zone indicates the twilight zone for which molecular modeling becomes increasingly difficult because sequence identity does not infer structural conservation. In the twilight zone, templates cannot be found with BLAST and threading methods must be used. When they fail, the dark zone is reached. Note that any two proteins have at least 5% sequence identity (dashed blue line).

Homology modeling is based on evolution. Proteins do not arise from scratch and can be classified into families. Homologous proteins within a family evolved from a common ancestor and share sequence, structural, and, to a lesser extent, functional similarities [7]. The structural conservation within a family is the keystone of molecular modeling. Using known structures of homologous proteins (the templates) and a multiple sequence alignment between the query and the templates, homology modeling programs such as MODELLER [8] build restraints and optimize the query structure from the template structures. In the safe zone, the sequence identity between the query and putative templates is high enough to infer that these proteins belong to the same family and share a common structure. By contrast, in the twilight zone, the similarities may arise from chance, convergence or common ancestry, which raises several issues:

(1) Finding templates by straightforward sequence-sequence comparison methods is not possible. This point has led to the development of “threading” methods based either on compatibility of sequence and fold or on profile-based search (see below).

(2) Additional information may be necessary to find evidence of common ancestry (homology) between template and query and infer the conservation of the structure. Indeed, very low identity rates may correspond to divergent or convergent evolution. In addition, random alignment leads to 5% sequence identity whereas, in highly divergent families, sequence identities can be as low as 8% (see an example in Fig. 2).

(3) Alignment between query and template(s) may be difficult. Indeed, the cutoff also separates easy from tricky alignments which may alter the quality of the homology modeling. Alignments are greatly improved by the use of multiple sequence alignment methods [9,10] but their accuracy remains challenging at low sequence identities.

Thus, the challenge of molecular modeling in the twilight zone relies on the recognition of correct templates and generation of accurate sequence-template alignments [6]. In this chapter, for clarity purpose, we will use, as a case study, the plethodontid receptivity factor isoform 1 (PRF1) from *Plethodon jordani*. This protein, for which structural and evolutionary data are missing, was discovered in 1999. It is a member of a pheromone protein family present only in lungless salamanders, with weak similarity with cytokines of the IL6 family [11]. We will show how sequence analysis methods, in particular multidimensional scaling (MDS), support the homology and structural conservation between PRF1 and cytokines of the IL6 family, by revealing intermediary sequences. We will also show how modeling of PRF1 can be improved by a variety of techniques.

2 Materials

2.1 Databases

1. UniProt (<https://www.uniprot.org/>) is a comprehensive resource of protein sequences and functional information [12]. It is composed of the manually curated SwissProt and of the automatically annotated trEMBL repositories. It contains not only protein sequences but also additional information including related 3D structures or models, and identifiers of the protein family in different family databases such as Pfam [13] and InterPro [14].
2. The Protein Data Bank (PDB, accessible at <https://www.rcsb.org/>) is the repository of biological macromolecular structures [2,15].
3. SCOP (Structural classification of proteins) [16] is a repository of protein folds, based on an initial classification into five structural classes: all alpha, all beta, alpha/beta, alpha+beta and small proteins.

2.2 Sequence analysis

1. NRDB90.pl [17] is a perl script aimed at clusterizing sequences based on sequence identity to build non-redundant sets. It can be downloaded from <ftp://biodisk.org>.
2. Different programs such as CLUSTAL [18], MUSCLE [19,20] and T-COFFEE [21] can be used to perform multiple sequence alignments (MSA).
3. The EXPRESSO program from the T-COFFEE suite [22] provides a multiple sequence alignment based on structural alignments, which may be a useful initial step for aligning proteins with low identities (<http://tcoffee.crg.cat/apps/tcoffee/do:expresso>).

4. MSA can be manually edited using Genedoc [23], a program aimed at editing and analyzing MSA through a graphical interface and available at <https://genedoc.software.informer.com/> . Subsequent phylogenetic analysis can be performed by the user-friendly MEGA software (Molecular Evolutionary Genetics Analysis) [24] available at <https://www.megasoftware.net/> .
5. The R package Bios2mds [25] is aimed at analyzing MSAs by multidimensional scaling and provides tools for user-friendly visualization of the sequence space (see Note 1).

2.3 Template mining

A variety of programs can be used to mine homologs in sequence databases. The choice of the program depends on the putative sequence identities between the query and the hits. Sequence - sequence comparison programs are adequate in the safe zone whereas sophisticated profile - profile searches are adapted to the twilight zone. Initially, “threading” referred to the search of a template by analyzing the compatibility of a sequence with a protein fold. Presently, “threading” refers to any method searching a template by sequence-profile or profile-profile comparison.

Here is a non-exhaustive list of sequence database mining programs:

1. BLAST (Basic Local Alignment Search Tool) [26], based on local sequence similarity, allows fast sequence-sequence comparison.
2. PSI-BLAST (Position-Specific Iterative BLAST) [27] is based on sequence-profile comparison. It derives a position-specific scoring matrix (PSSM) from the multiple sequence alignment of sequences detected above a given score threshold using protein-protein BLAST.
3. HMMER [28] is a sequence-profile comparison method based on profile hidden Markov models (HMMs) (<https://toolkit.tuebingen.mpg.de/tools/hmmer>).
4. Phyre2 (Protein Homology/analogY Recognition Engine V 2.0) [29] performs its searches by mining a database of profile HMMs, one for each known 3D structure (<http://www.sbg.bio.ic.ac.uk/~phyre2/>).
5. HHPred [30-32] performs HMM-HMM profile searches in sequence databases to find homologs (<https://toolkit.tuebingen.mpg.de/tools/hhpred>).
6. LOMETS (local meta-threading-server) [33] performs template searches using eleven different threading methods (see Note 2). Starting from a query sequence, LOMETS works in three steps: (1) Building a sensitive (or deep) MSA, (2) Threading the deep MSA by individual programs, and (3) Ranking templates with a specific scoring function which takes into account normalized Z-scores and sequence identities. For each method, the normalized Z-scores differentiate good/bad templates (threshold of 1) (<https://zhanglab.ccmb.med.umich.edu/LOMETS/>).
7. SUPERFAMILY finds a protein fold based on a collection of hidden Markov models, which represent structural protein domains at the SCOP superfamily level [34,35]. SUPERFAMILY is a “true” threading program, aimed at finding a protein fold (<https://supfam.org/SUPERFAMILY/>).

2.4 Secondary structure prediction

When searching information for a protein without close structurally resolved homologs, prediction of secondary structure (SS) may yield useful information. Best performances for SS prediction are obtained with programs based on multiple sequence alignment profiles and neural networks, such as PSIPRED [36,37] (<http://bioinf.cs.ucl.ac.uk/psipred/>) and SPIDER3 [38] (<https://sparks-lab.org/server/spider3/>).

2.5 Automated structure prediction

Several web servers can perform automated structural prediction from a query sequence. They are based on threading methods to find templates, and then they use different methods for the subsequent modeling steps. Here, we list three automated 3D prediction servers compared in this Chapter:

1. Phyre2 [29] : After template detection by mining a HMM database, the subsequent molecular modeling step is carried out with MODELLER based on the resulting sequence alignment between query and templates (<http://www.sbg.bio.ic.ac.uk/~phyre2/>).
2. I-TASSER (Iterative Threading ASSEmbly Refinement) [39-41] is a hierarchical approach to protein structure prediction. It first identifies structural templates by comparing the best hits from the “best” ten out of fourteen threading methods with LOMETS, and then it builds full-length atomic models by iterative template-based fragment assembly simulations (<https://zhanglab.ccmb.med.umich.edu/I-TASSER/>).
3. ROBETTA [42] is a protein structure prediction service. In the option for Rosetta Comparative Modeling (RosettaCM) [42], four independent methods (see Note 3) are used to detect templates and generate sequence alignments, and then models are built from template hybridization (<https://rosetta.bakerlab.org>).

2.6 Customized structure prediction

For users who wish to build their own models, the MODELLER program [8] builds molecular models of the query from the template structure(s) by minimizing structural, stereochemical and user-defined restraints. The structural restraints are based on the structure of the template(s) and the alignment between query and template(s). To customize models in order to match structural and functional requirements, expert information can be introduced by (1) adding user-defined restraints such as distance between two residues or secondary structure elements in the modeling procedure and (2) by combining user-selected templates and template fragments.

2.7 Model validation

With threading methods, the metrics to compare models and templates must be more sensitive to the global fold similarity than to local structural variation. This is not the case of the traditional root-mean-square deviation (RMSD). The TM-score (see Note 4) has been specifically designed to solve this problem [43]. A threshold of 0.5 differentiates proteins with similar fold from proteins with different fold [44]. TM-scores can be calculated after structural alignment with TM-align [45] at the Zhang lab server (<https://zhanglab.ccmb.med.umich.edu/TM-align/>).

2.8 Graphical analysis

Graphical analysis of templates and models can be carried out by a variety of molecular visualization programs, such as Chimera [46] or Pymol (<https://pymol.org/>). Note that the low identity rates between template and query sequences in the twilight zone prevent the use of sequence-based structural superposition functions, such as the align function in Pymol.

3 Methods

3.1 Our case study

As a case study, we chose a small protein, the plethodontid receptivity factor isoform 1 (PRF1) from *Plethodon jordani* (UniProt entry: Q9PUJ2_PLEJO) [11]. This 215 amino acid protein (including a 23 amino acid peptide signal) is a courtship pheromone produced by males to increase female receptivity (see Note 5). When PRF1 was discovered in 1999, it was acknowledged that its sequence was weakly related (around 16% sequence identity) to the ciliary neurotrophic factor (CNTF) and cardiotrophin-1 (CTF1), two cytokines of the interleukin-6 (IL6) family [11]. Since then, two additional cytokines of the IL6 family have been discovered: cardiotrophin-2 (CTF2, absent in humans) [47] and the cardiotrophin-like cytokine factor 1 (CLCF1) [48]. CTF2 and CLCF1 have, respectively, 26% and 20% sequence identity with PRF1.

In mammals, the IL6 family of cytokines includes IL6, interleukin-11 (IL11), leukemia Inhibitory factor (LIF), oncostatin M (ONCM), CNTF, CTF1, CTF2 and CLCF1. Albeit the sequence identities can be as low as 8%, these cytokines share a common four-helix bundle fold with an up-up-down-down topology (Fig. 2). In addition, they signal through the gp130 receptor subunit and share similar binding sites with cognate receptors (see Note 6) [49-52]. Crystal structures have been resolved for five cytokines from the IL6 family: IL6 (1ALU [53], 5FUC [54]), IL11 (4MHL [55]), CNTF (1CNT [56]), LIF (1EMR, 1LKI [57], 2Q7N [58], and ONCM (1EVS [59]). No crystal or NMR structure has been reported to date for PRF1 or cardiotrophin-like cytokines.

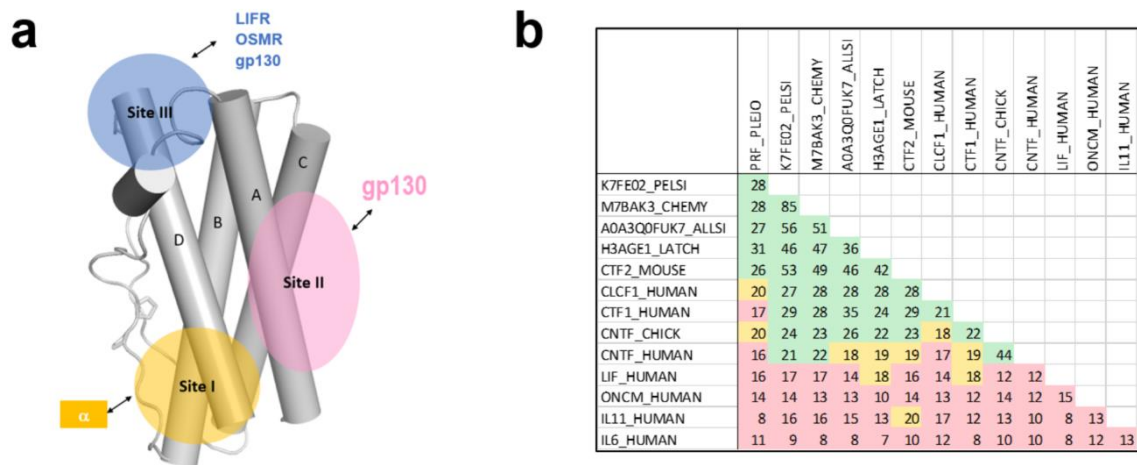


Fig. 2: Cytokines of the IL6 family. (a) General up-up-down-down topology of the four-helix bundle fold of this cytokine family. The helices are numbered from A to D. The positions of the three conserved sites of interaction with the cognate receptors are indicated. Site II interacts with gp130, site III interacts with either gp130, LIFR or OSMR, while site I can interact with a third, specific, “ α ” receptor. See Note 6 for details; (b) Sequence identities between PRF1, its closest homologs and the cytokines of the IL6 family. The color code indicates the reliability of the sequence identity (green: safe zone, yellow: transition zone, red: twilight zone).

3.2 Template search by PDB mining

In InterPro, PRF1 is described as belonging to the *4_helix_cytokine-like_core* superfamily (IPR009079) and to the *PRF/Cardiotrophin-like* family (IPR010681). Additional information from the PRF1 sequence was searched for using the SUPERFAMILY assignment server [34]. SUPERFAMILY predicts that PRF1 is in the class of *All alpha proteins*, and belongs to the fold/superfamily of *4-helical*

cytokines with an E-value of 3×10^{-55} (see Note 7). It also suggests an “uncertain” classification for the family level as *Long-chain cytokine* with an E-value of only 0.005.

The next step to find a template was the mining of the PDB in search of homologs. Using the mature sequence of PRF1 (residues 24-215) as a query, we performed different searches (Table 1):

1. Straightforward mining of the PDB with BLAST: this search led to no hit
2. Sequence - profile search: A PSI-BLAST search seeded with the PRF1 sequence, followed by selection of hits on most query sequence (>60%), led to CNTF as a hit with an unreliable E-value of 5.9
3. Profile HMM search: Profile search using the HMMER program [28] was carried out on the HHpred server. The search led to two hits, CNTF and LIF, as putative templates with very significant E-values of 10^{-10} or lower
4. Profile HMM - profile HMM search: the HHpred algorithm led to six hits with E-values lower than 0.1: CNTF, LIF, ONCM, IL11, GCSF (granulocyte colony-stimulating factor) and IL6. Among them, GCSF (E-value of 8×10^{-13}) is a 4-helical cytokine that does not belong to the IL6 family, but shares the same up-up-down-down four-helix bundle fold.

TABLE 1
PDB mining using the PRF1 sequence as a query

Search method	Program	Hits ¹	E-value
Sequence based	BLAST	No hit	
Sequence profile based	PSI-BLAST	CNTF	5.9
Profile HMM based	HMMER	CNTF LIF	7×10^{-20} 4×10^{-10}
Profile HMM – Profile HMM based	HHpred	CNTF LIF IL11 ONCM <i>GCSK</i> IL6	7×10^{-32} 3×10^{-31} 5×10^{-28} 8×10^{-26} 8×10^{-13} 2×10^{-7}

¹ For clarity purpose, only the proteins (and not the PDB numbers) are indicated. Italics fonts indicate a growth factor with the same four-helix bundle fold as the IL6 family.

3.3 Template search with LOMETS

Finally, a comparison of 11 threading methods was carried out with the LOMETS server [33] (Table 2). All the methods, but one, classified CNTF or LIF as best hits. Only CEthreader, which is a contact-based method, privileged prolactin (PDB 1RW5). This growth factor shares the four-helix bundle fold of the IL6 cytokines. Most additional hits include cytokines of the IL6 family (ONCM, IL11, IL6) or cytokines/growth factors with same four-helix bundle fold (prolactin, lactogen, IL23). However, several methods also found IL1Ra, the interleukin-1 receptor antagonist (PDB 1ILR) [60] that has a beta barrel fold (see Note 8). This finding serves as a reminder of how cautious users need to be when analyzing threading results.

TABLE 2

Comparison of the LOMETS threading programs using the PRF1 sequence as a query

Program ¹	Method ²	Top hit ³	Additional hits with $Z_n > 1$ ^{3,4}
HHpred	HMM	CNTF	LIF, IL11, ONCM
CEthreader	Contact	<i>Prolactin</i>	IL11, CNTF, <i>lactogen</i> , LIF, IL1Ra
SparksX	Profile	CNTF	LIF, IL11, ONCM, IL1Ra , IL6, <i>prolactin</i>
FFAS3D	Profile	CNTF	LIF, ONCM, IL11, <i>GCSF</i> , IL6, IL1Ra , <i>prolactin</i> , <i>lactogen</i>
MUSTER	Profile	LIF	CNTF, ONCM, IL11
Neff-MUSTER	Profile	LIF	CNTF
HHsearch	HMM	LIF	CNTF, ONCM, IL11, <i>GCSF</i> , IL6, IL1Ra
SP3	Profile	LIF	CNTF, ONCM, IL11, IL6, <i>GCSF</i> , IL1Ra , <i>prolactin</i>
PPAS	Profile	CNTF	LIF, ONCM, IL11, <i>GCSF</i> , IL6, IL1Ra , <i>IL23</i>
PROSPECTOR2	Profile	CNTF	LIF, ONCM, IL11
PRC	HMM	LIF	CNTF, ONCM, IL11

¹ The programs are ranked as determined by LOMETS. See Note 2 for references.

² HMM corresponds to profile HMM – profile HMM based searches; profile corresponds to sequence profile – sequence profile based searches.

³ For clarity purpose, only the proteins (and not the PDB numbers) are indicated.

⁴ Normalized Z-scores (Z_n) indicate the quality of the hits. They are considered “good” above the threshold of 1. The hits are sorted by decreasing Z_n . When several hits correspond to the same protein (different origins, conditions or methods), only the first hit is indicated. Italics fonts indicate four-helix bundle cytokines/growth factors that do not belong to the IL6 family but share the same fold. The bold fonts for IL1Ra highlight a hit with a beta barrel fold.

3.4 Sequence space investigation

As exemplified by IL1Ra in LOMETS results (Table 2), finding a template by threading does not prove that there is homology, i.e. a common ancestor, between the template and the query, nor that the structure is conserved. The identity of 16% between the PRF1 query and the CNTF or LIF templates is positioned in the twilight zone (Fig. 1). However, homology and structural conservation are transitive properties. Investigation of the sequence space of the query analogs may reveal sequences homologous to both query and template(s) in the safe zone and consequently validate the template. Indeed, finding intermediates is an efficient strategy to reduce false positives [6,30].

To investigate the query sequence space, several steps must be carried out:

1. Blast search of the query homologs in sequence databases. Here, using PRF1 as a query in UniProt vertebrate sequences, we obtained 602 hits with E-value lower than 10. Among them, 190 sequences corresponded to salamander receptivity factors and shares with PRF1 sequence identities larger than 60%. Among very significant hits ($E\text{-value} < 10^{-10}$), several sequences are known cytokines: human and mouse CLCF1, human and mouse CTF1, mouse CTF2. Interestingly, the sequence of chicken CNTF was identified with an E-value of 5 and a sequence identity of 20%.
2. Building of a non-redundant (NR) set of the hits. This step considerably reduces the number of sequences investigated by suppressing non informative, highly similar sequences. Here, we built a NR set of PRF1 hits (id <90%), first by selecting sequences with length between 150 and 360 aa, and then, by clustering these sequences using the NRDB90.pl script [17]. The MSA of the NR set

was carried with Clustal [18] and manually edited with Genedoc [23]. At this stage, a few additional truncated sequences were removed. This led to a non-redundant set of 125 aligned sequence sequences to which we added the human CNTF sequence. Computation of sequence identities on this alignment revealed that the closest neighbors to PRF1 are proteins from coelacanth (H3AGE1_LATCH, 31% id), turtle (K7FE02_PELSI, M7BAK3_CHEMY, 28%) and alligator (A0A3Q0FUK7_ALLSI, 27%).

3. Building a phylogenetic tree of the NR set. Using the alignment of the NR set, a Neighbor Joining tree was built with the MEGA software [24] (Fig. 3a). This tree visualizes the PRF, CTF1, CTF2, CLCF1 and CNTF sub-families along with the closest neighbors of PRF1.
4. Visualizing the sequence space of the non-redundant set. Using the bios2mds R package [25], we could visualize the sequence space of the PRF1 homologs by multidimensional scaling (Fig. 3b). The first two dimensions are driven by the differences between CTF1, CTF2 and CLCF1 cytokines. CNTFs and PRFs are projected towards the center of the first two dimensions. To better visualize the relationship between these proteins, we projected the sequences onto the third and fourth dimensions. In this case, we observed the closest neighbors of PRF1 at intermediary positions between PRFs and CNTFs. Indeed, these sequences are above the twilight threshold for human and/or chicken CNTF, strongly supporting the assumption of homology between PRF1 and CNTF (Fig. 2b).

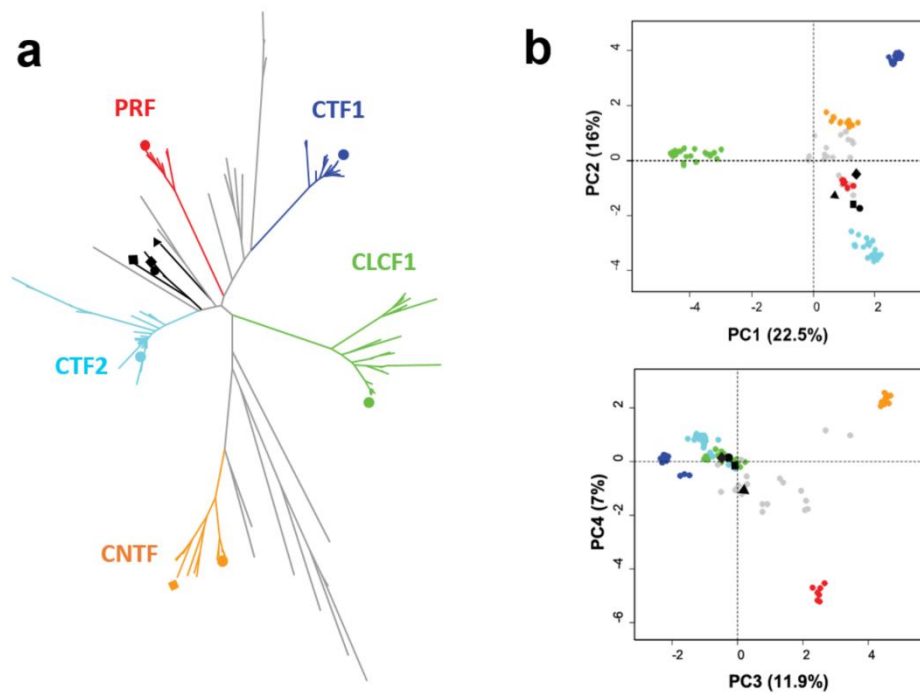


Fig. 3: Evolutionary information on PRF1. (a) NJ tree of PRF1 homologs (500 bootstraps); (b) Sequence space of the PRF1 homologs with projection of the sequences onto the first and second dimensions (top) and onto the third and fourth dimensions (bottom) of the MDS analysis. The PRF1 homologs were obtained by mining UniProt with BLAST, followed by clustering to obtain a non-redundant set. The MDS analysis was carried out with the bios2mds package. In (a) and (b), the color code is as follows: PRF proteins: red, CNTF: orange, CTF1: dark blue, CTF2: magenta, CLCF1: green, the closest four homologs: black, others: grey. In (a), labels indicate PRF1 (red circle), human CTF1 (blue circle), murine CTF2 (cyan circle), human CLCF1 (green circle), human CNTF (orange diamond), chicken CNTF (orange circle), K7FE02_PELSI (black circle), M7BAK3_CHEMY (black square), H3AGE1_LATCH (black triangle) and A0A3Q0FUQ7_ALLSI (black diamond). In (b), only black labels for closest PRF1 homologs are shown.

3.5 Template analysis

As several structures of IL6 cytokines are available, careful graphical analysis may give information on conserved and variable elements that should be taken into account for modeling. Models are less reliable in the variable parts and more reliable in variable parts.

Fig. 4a displays the crystal structure of human CNTF superimposed with structure of LIF, IL6, ONCM and IL11. The superposition was carried out with the super function of Pymol, based on structures because the low sequence identities between these cytokines prevented the use of the align function based on sequence identities.

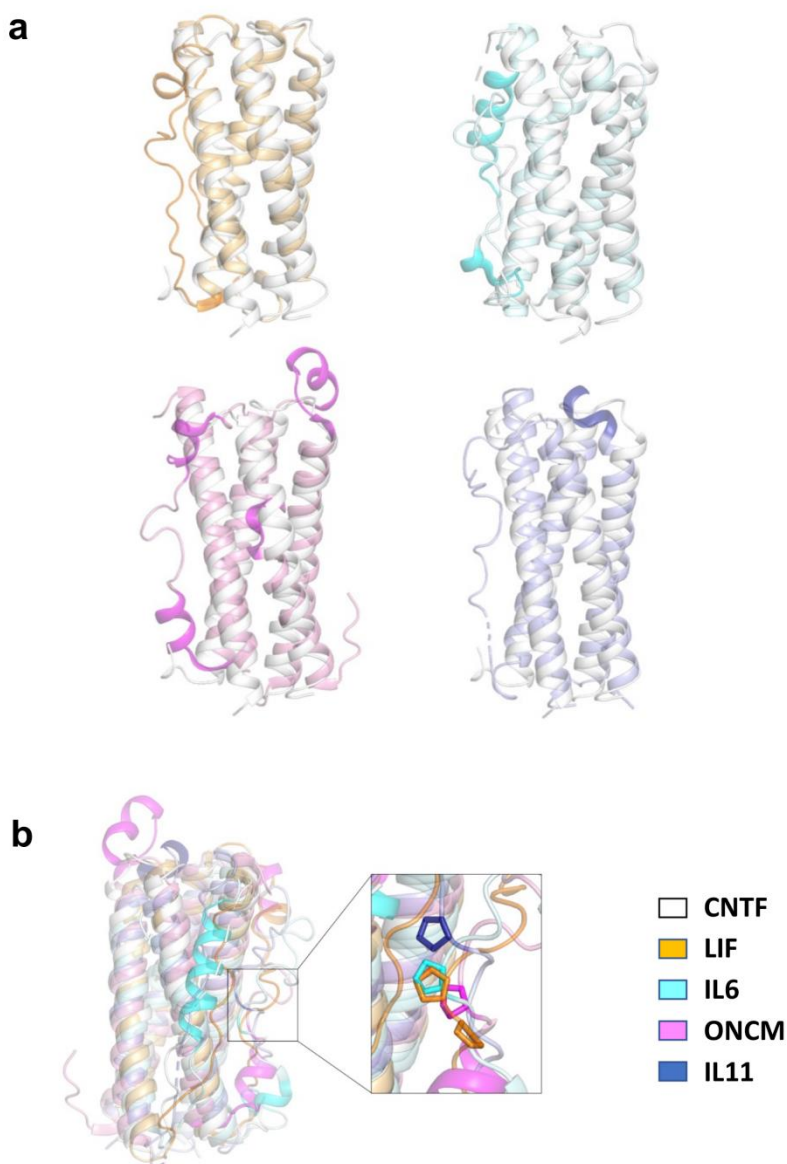


Fig. 4: Comparison of different templates. In (a), a ribbon representation of CNTF (white) is superimposed with LIF (orange), IL6 (cyan), ONCM (magenta) and IL11 (blue). Regions structurally different as compared to CNTF are indicated by a darker color for clarity purpose. In (b), zooming on the AB loop reveals the conserved positioning of a proline residue (shown as stick) in the cleft between helices B and D in LIF, IL6, ONCM and IL11. Note that the loop AB is absent in the CNTF structure. The PDB files are 1CNT (CNTF), 1EMR (LIF), 5FUC (IL6), 1EVS (ONCM) and 4MHL (IL11).

The general topology of the four-helix bundle is conserved but specific properties are observed. LIF has shorter helices A and D on the “bottom” side but otherwise does not present marked differences. Shorter helices A and D are also observed for IL6. In addition, IL6 has shorter helices B and C on the “top” side and a 12 residue long α -helix in loop CD. ONCM has a strong distortion in helix A, an insertion with two helical turns in loop BC and two helical elements in loop AB. A helical element is also observed in loop BC of IL11 and distorts the C-terminus of helix B.

3.6 Sequence alignment

MODELLER uses spatial restraints based on the alignment between query and template(s) to build a model. The alignment step is thus crucial for the quality of the model. In the twilight zone, straightforward alignment of the query and the template should be avoided as they do not lead to reliable alignment. It was early acknowledged that MSA methods lead to improved quality of the alignment [9]. Here are some tips:

1. First, use large homolog sets to build an MSA. Then, manually inspect the MSA, compare it to structural alignment of the templates and correct it if necessary with Genedoc [23]. This should limit template/query alignment errors.
2. You may use EXPRESSO [22] to obtain an initial sequence alignment of all the putative templates based on their structural alignment. Compare the sequence alignment to the superimposed structures and correct if necessary.
3. Subsequently, you may use this template profile as a seed to align the query and its homologs.
4. Visual inspection of MSA is mandatory to insure that the alignment is correct.
5. In any case, never forget that sequence alignment may be the limiting step of homology modeling.

3.7 Loop modeling

Long loops are challenging to reliably model due to combination of different factors: (1) they have high sequence and structural variability, (2) they are frequently missing in templates (e.g. the CNTF template in which loops AB and CD are missing and partly missing, respectively), and (3) the functions developed for loop modeling, such as `model_loop` in MODELLER [61], are suited to short loops but not long loops. Two options are possible to improve modeling of long loops: (1) SS prediction programs may suggest SS elements in these long loops that can be introduced in the modeling procedure; and (2) loops with the same (or, if not possible, similar) size in related proteins may be used as templates.

In our case study, the long loops joining helices A and B, and helices C and D are highly variable and they are missing or partially missing in all the templates except LIF (Fig. 5). However, we note that the loops AB of LIF, ONCM and IL6 present helical elements. In loops CD, a long helix is observed only for IL6. To gain information on putative secondary elements in the long loops of PRF1, we used the SS prediction programs PSIPRED [36] and SPIDER3 [38]. Both programs predicted a helical element at the C-terminus of loop AB that should be taken into account for modeling (Fig. 5). In addition, careful scrutiny of the templates indicates the strong conservation of a proline residue from loop AB at the cleft between helices B and D (Fig. 4b). This position might tether the loop AB at the cytokine surface and should be an element to take into account for PRF1 modeling.

Fig. 6 displays the three automated models superimposed on the customized MODELLER model. No significant difference was observed for the four-helix bundle core, but the loops AB were markedly different. Among automated models, only the Phyre2 model predicted a position of Pro-61 similar to the customized model. By contrast, only RosettaCM predicted a helical element at the C-terminus of loop AB. However, the exposed orientation of Phe-70, similar to the orientation of Trp-64 in CNTF, was observed only in the customized model, supporting the positioning of the helical element predicted by SPIDER3 and PSIPRED.

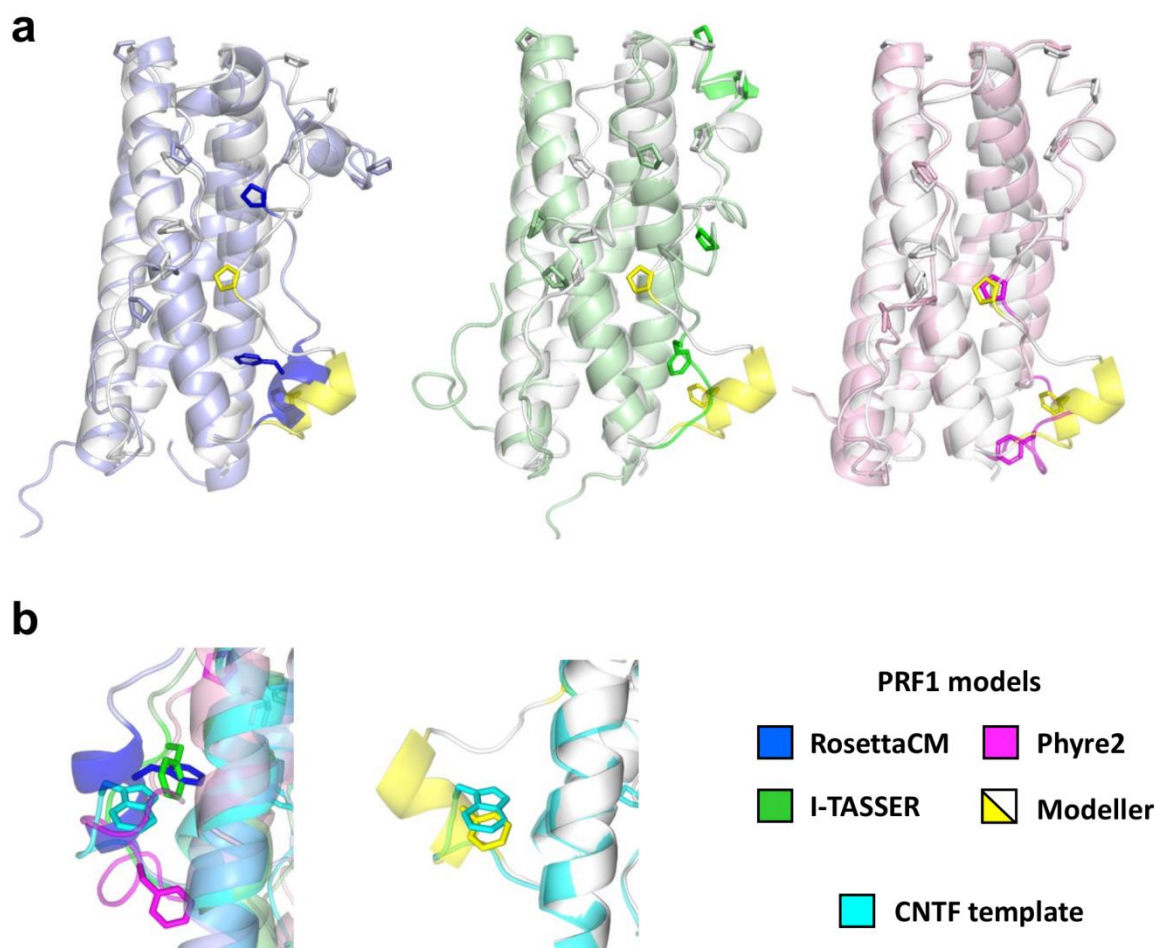


Fig. 6: Comparison of the PRF1 models. In (a), the customized MODELLER model (white and yellow) is superimposed with the best models obtained with RosettaCM (blue), I-TASSER (green) and Phyre2 (pink). Pro-61 and Phe-70 are indicated by sticks. The yellow ribbon in the MODELLER model indicates the user-added helical restrains. In (b), the zooming on the C-terminal part of loop AB reveals that the superposition of the phenyl ring of CNTF Trp-64 (cyan) with PRF1 Phe-70 (yellow) in the customized MODELLER model is not observed in the other three models.

The TM-scores between models and templates (Table 3) can assess the quality of the models. The higher scores obtained by the customized and the Phyre2 models with the CNTF and LIF templates, respectively, reflect the templates favored by MODELLER in the modelling procedure, while the I-TASSER and RosettaCM work by template based fragment assembly or model hybridization, and do not match a template as closely as the customized and Phyre2 models. In any case, the high TM-scores are consistent with similar folds and corroborate the “threadability” of PRF1 [65].

TABLE 3

TM-scores¹ of the PRF1 models

Method	CNTF	LIF
I-TASSER	0.71	0.76
RosettaCM	0.69	0.69
Phyre2	0.68	0.88
Customized	0.78	0.85

¹ The TM-scores between the PRF1 models and the CNTF and LIF templates were computed using the length of PRF1 as reference.

3.9 Concluding remarks

This chapter provides an example of homology modeling in the twilight zone, using conventional methods. It is important to note that all the threading methods used to find templates led to similar results (Tables 1 and 2). This is a general observation that threading methods succeed or fail together for molecular modeling of proteins in the twilight zone [65]. Indeed, “best” templates of any method, including CEthreader, share the up-up-down-down four-helix bundle fold of the IL6 family. Nevertheless, several methods found a hit with a beta barrel fold. Caution is thus always recommended when threading methods must be used. Here, the TM-scores obtained for the PRF1 models (Table 3) indicate high “threadability”, which is consistent with the four-helix bundle fold recognized as “best” hit by the different threading methods. The sequence space analysis corroborates the homology between the query and the templates by finding intermediate sequences linking PRF1 and CNTF. The most difficult task remains the modeling of the long loops. In this case, additional information provided by the user or by deep learning, based on structural or functional criteria, may improve the quality of the model. While this chapter was in press, the PRF1 model obtained with AlphaFold was released (AF-Q9PUJ2-F1, available on Uniprot). Notably, this model predicted the same alpha-helical structure at the C-terminus of the loop AB (rmsd of 1.7 Å), with the same *trans* orientation of Phe-70 towards helix D, as our customized model (Supplementary Fig. S1). Thus, AlphaFold was able to automatically mine and integrate additional information (here secondary structure predictions) in its modeling procedure, as we have done manually. Finally, “unthreadability” appears as a property inherent to protein fold [65]. Exploration of the dark zone will require the development of *ab initio* approaches to deduce the three dimensional structure of a protein from physics and physico-chemical principles alone.

4 Notes

1. Multidimensional scaling is a multivariate analysis method that transforms a distance matrix into points in a low dimensional space. The distances between points in the resulting space are as close as possible to the distances in the original matrix [66]. When applied to an MSA, this method allows the 2D or 3D visualization of the sequences (the sequence space) based on the identity or on the similarity matrix of the sequences in the MSA.
2. LOMETS [33] compares eleven threading methods. Four of these methods (CEthreader, MUSTER, Neff-MUSTER and PPAS) have been developed by Zhang and coworkers [33,67,68]. The methods are based on contact searches (CEthreader), sequence profile – sequence profile searches

(MUSTER, Neff-MUSTER, FFAS-3D [69], PPAS, PROSPECT2 [70], SP3 [71], SparksX [72]) and profile HMM – profile HMM searches (HHsearch [31], HHpred [30], and PRC [73]).

3. In RosettaCM, the four independent threading methods used to detect templates are RaptorX [74], HHpred [30], Sparks-X [72], and Map align [75].
4. TM-score is a metric for assessing the topological similarity of protein structures. It gives more weight to smaller deviations than to larger deviations and is thus more sensitive to the global fold than traditional root mean square deviations. In addition, it is scaled by sequence length. TM-scores have values in [0,1] range, with 1 indicating a perfect match between two structures. Scores are higher than 0.5 for structures with similar folds and lower than 0.17 for randomly chosen unrelated proteins [41,43]. Interestingly, a TM-score threshold of 0.4 can also be used to differentiate “threadable” from “non threadable” proteins. The “threadability” of a protein appears related to inherent properties of the protein fold [42].
5. The *Plethodontidae* or lungless salamanders have complex mating behaviors and courtship rituals. PRF1 is produced in a gland on the male’s chin and delivered to the female during courtship by scratching or slapping. A male using the scratching mode of delivery administers the pheromone by wiping his mental gland across the female’s dorsum while scraping her skin to deliver the pheromone directly into the female’s circulatory system. In the slapping delivery mode, the male slaps its gland directly across the female’s nares to deliver the pheromone to chemoreceptors in the vomeronasal organ [76].
6. The cytokines of the IL6 family [49-51] are long chain four-helix bundle proteins. In spite of low sequence identities (8-20%), these cytokines share a common fold with an up-up-down-down topology for the four A to D helices. In mammals, the family includes IL6, IL11, LIF, ONCM, CNTF, CTF1, CTF2 (absent in humans) and CLCF1. These cytokines recruit two receptors, including the common gp130 receptor and a second variable receptor (gp130, LIFR or OSMR). The two receptors transactivate each other to trigger a cellular response through JAK/STAT pathways. In addition, a third receptor (the α receptor) without intracellular part may be recruited to increase the affinity. These receptors bind three conserved sites of the cytokines (Fig. 2). The site I located on helix D and loop AB binds the α receptor, site II located on helices A and C binds gp130 and site III at the top of helix D binds either gp130, LIFR or OSMR. The signature of LIFR binding is an FxxK motif at the N-terminus of helix D.
7. Expect-value or E-value describes the number of hits “expected” by chance when searching a database of a particular size. It depends on both the matching score and the length of the query. The closer to 0 the E-value is, the more significant the hit is.
8. Six threading programs (CEthreader, SparksX, FFAS3D, HHsearch, SP3, PPAS), based on three different search methods, find the interleukin-1 receptor antagonist (IL1Ra) as a putative “good” hit. This protein (PDB: 1ILR) has a very different beta trefoil structure [60]. The beta trefoil fold consists of six beta hairpins, each formed by two beta strands. Together they form a beta barrel with a triangular cap and an approximate three-fold symmetry.

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Supplementary Figure 1

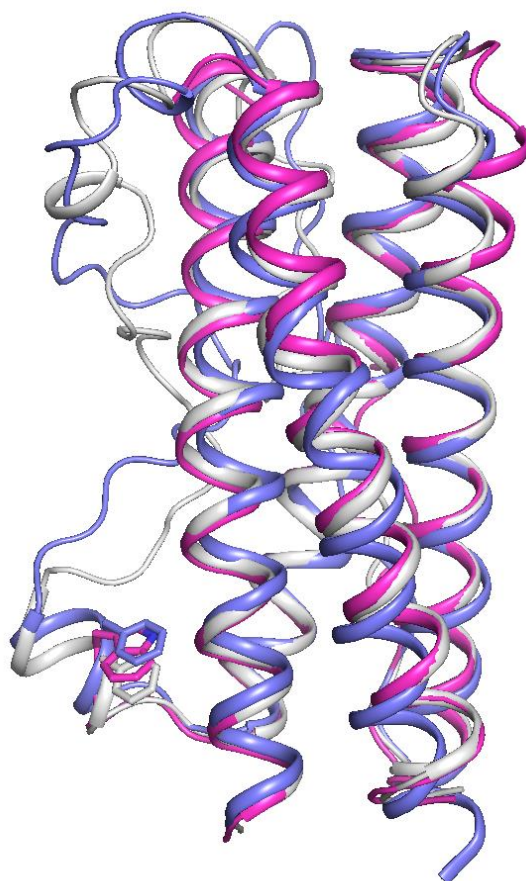


Fig S1: Comparison of the customized and AlphaFold models. The customized MODELLER model (residues 10-187 of mature protein, white ribbon) is superimposed with the AlphaFold model deposited in UniProt (AF_Q9PUJ2-F1, slate) and with the CNTF template (PDB access number: 1CNT, magenta). Phe-70 of PRF1 and Trp-64 of CNTF are shown as sticks.